**Project Design Phase-II**
**TechnologyStack(Architecture&Stack)**

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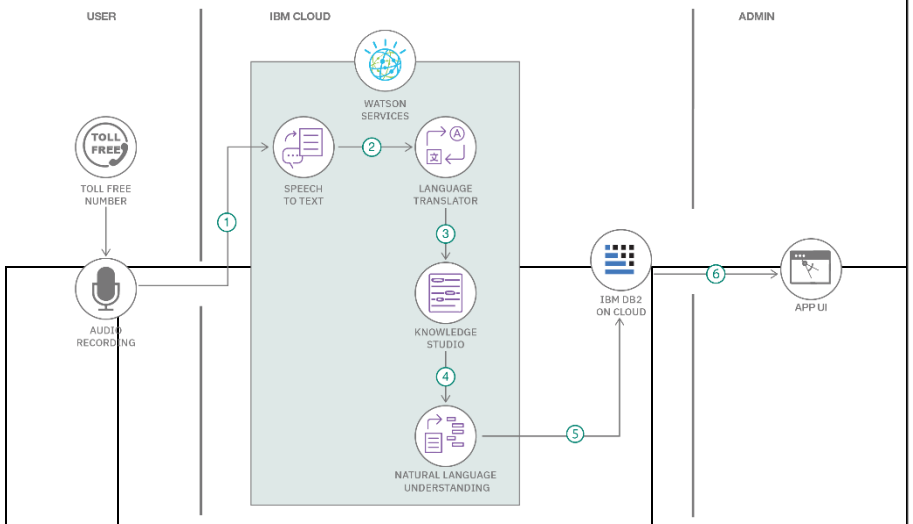
|---|---|
| Date | 09 March 2026 |
| ProjectName | UCab – Cab Booking Web Application |

**TechnicalArchitecture:**

TheDeliverableshallincludethearchitecturaldiagramasbelowandtheinformationasperthetable1&table2

**Example:Orderprocessingduringpandemicsforofflinemode**

**Reference:**[https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/](https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/)




Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

**Table-1:Components&Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | Interface for users and admins to interact with the system | React.js, HTML, CSS, Bootstrap |
| 2 | Application Logic 1 | Backend logic for authentication and API handling | Node.js, Express.js |
| 3 | Application Logic 2 | Business logic for booking and fare calculation | JavaScript (Node.js) |
| 4 | Application Logic 3 | Admin management and cab management logic | Express.js controllers |

| | | | |
|---|---|---|---|
| 5 | Database | Stores users, cabs, bookings, and admin data | MongoDB |
| 6 | Cloud Database | Optional cloud database deployment | MongoDB Atlas |
| 7 | File Storage | Storage for uploaded cab images | Local server / Cloud storage |
| 8 | External API 1 | Future integration for maps and distance calculation | Google Maps API |
| 9 | External API 2 | Future integration for online payment gateway | Razorpay / Stripe |
| 10 | Machine Learning Model | Future ride demand prediction | Python ML model (optional) |
| 11 | Infrastructure | Deployment environment for the application | Local server / AWS / Render |

**Table-2:ApplicationCharacteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-SourceFrameworks | Listtheopen-sourceframeworksused | TechnologyofOpensourceframework |
| 2. | SecurityImplementations | Listallthesecurity/accesscontrolsimplemented, use of firewalls etc. | e.g.SHA-256,Encryptions,IAM Controls, OWASP etc. |
| 3. | ScalableArchitecture | Justifythescalabilityofarchitecture(3–tier, Micro-services) | Technologyused |
| 4. | Availability | Justifytheavailabilityofapplication(e.g.useof load balancers, distributed servers etc.) | Technologyused |

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 5. | Performance | Designconsiderationfortheperformanceofthe application (number of requests per sec, use of Cache, use of CDN's) etc. | Technologyused |

**References:**


https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-

pandemic/https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d